Solution Design Document

Order Processing Modernisation Programme

Prepared for: Lumen Home Goods (the customer) | Prepared by: Brightpath Cloud Consulting | Version 1.0 FINAL

1. Executive Summary

Lumen Home Goods is a mid-sized online retailer of furniture and home accessories. Over the last two years the customer has seen order volumes grow by roughly 140 percent, and the existing monolithic order system, hosted on a single rented server, can no longer absorb seasonal peaks without manual intervention. This document describes the finalised solution that Brightpath Cloud Consulting will deliver to replace that system with a serverless, event-driven order processing platform on Amazon Web Services.

The programme is sponsored by the customer's Head of Digital and is expected to reduce checkout failures during peak periods, shorten the time from order placement to confirmation email, and remove the operational burden of patching and capacity planning. The design has been reviewed with the customer's engineering lead and the finance team and is considered final for the purposes of the delivery plan and commercial terms in this document.

2. Business Context and Objectives

Black Friday and the pre-holiday period account for nearly 45 percent of the customer's annual revenue. During the last peak, roughly one in twenty checkout attempts timed out. The customer has set four objectives for the programme: sustain ten times the current peak order rate without degradation, confirm at least 99 percent of valid orders within 30 seconds, keep the cost per order flat as volume grows, and give the customer's support staff a reliable record of the state of every order.

Success will be measured by the checkout completion rate, the median and 99th percentile order-confirmation latency, and the number of manual interventions per thousand orders. Baselines were captured during discovery workshops held in March and April.

3. Scope

In scope are the customer-facing storefront web application, customer sign-up and sign-in, the order intake API, asynchronous order processing including inventory reservation and payment, and transactional email notifications. Out of scope are native mobile applications, replacement of the customer's ERP and warehouse management systems, migration of historical orders older than 24 months, and any change to the product catalogue management process. The customer will remain responsible for its agreement with the external payment provider.

4. Proposed Solution Architecture

The platform is deployed in a single AWS Region, Europe (Ireland) eu-west-1. It follows a serverless pattern throughout: there are no servers, containers or virtual networks for the delivery team to operate. The remainder of this section walks through the solution from the shopper's browser to the back-end workflow.

4.1 Storefront delivery

Shoppers reach the storefront through their web browser. The storefront is a static single-page application whose build artefacts are stored in a private Amazon S3 bucket. Amazon CloudFront is placed in front of the bucket and serves the application to shoppers worldwide; CloudFront reads from the bucket through an origin access control so that the bucket itself is never publicly readable. The same CloudFront distribution also fronts the API so that the browser talks to a single domain.

4.2 Identity

Customer accounts are managed by Amazon Cognito. A Cognito user pool handles registration, sign-in, password recovery and multi-factor authentication. After signing in, the browser application receives a JSON Web Token from Amazon Cognito and presents it on every API call.

4.3 Order intake

The browser calls a REST API exposed by Amazon API Gateway. API Gateway validates the token with a Cognito authorizer and invokes an AWS Lambda function named the Order API function. This function validates the basket, writes a new order record with the status PENDING to an Amazon DynamoDB table called Orders, and then places an order message on an Amazon SQS queue. A dead-letter queue captures messages that fail processing after three attempts. The function returns an order identifier to the browser immediately, so the shopper never waits on payment processing.

4.4 Asynchronous processing and orchestration

A second AWS Lambda function, the Order Processor, consumes messages from the Amazon SQS queue and starts one AWS Step Functions Standard workflow execution per order. The workflow performs the business steps in a fixed order. First it reserves stock by updating an Inventory table in Amazon DynamoDB using the native Step Functions service integration. Second, it invokes a Payment Lambda function that calls the HTTPS API of PayFlow, the external payment provider, to authorise and capture the payment. Third, it updates the order record in the Orders table to CONFIRMED. Finally, it publishes an OrderConfirmed event to a custom Amazon EventBridge event bus. If the payment is declined, the workflow releases the reserved stock and sets the order status to FAILED.

4.5 Notifications and events

An Amazon EventBridge rule matches OrderConfirmed events and invokes a Notification Lambda function. That function renders the confirmation email and sends it to the shopper through Amazon SES. Using Amazon EventBridge keeps the workflow decoupled from notifications, so that additional consumers, such as a future loyalty-points service, can subscribe later without any change to the order workflow.

4.6 Observability and security

All Lambda functions and the workflow emit logs and metrics to Amazon CloudWatch, and the customer's support staff will use a CloudWatch dashboard and alarms on queue depth and workflow failures. Each Lambda function runs with its own least-privilege IAM role, and the PayFlow API credentials are held in AWS Secrets Manager and read by the Payment Lambda function at runtime.

5. Delivery Timeline

Phase	Duration	Key outputs
Mobilisation and design sign-off	2 weeks	Final solution design, delivery plan
Platform foundation and storefront	4 weeks	Static site, identity, CI/CD pipelines
Order intake and processing	6 weeks	API, queue, workflow, payment integration
Notifications, testing and cut-over	4 weeks	Email, load tests, go-live
Hypercare	3 weeks	Daily review, tuning, handover

The total elapsed duration is 19 weeks from the start of mobilisation. Dates will be fixed once the customer confirms the start date and nominates its delivery team.

6. Commercial Assumptions

The engagement is priced on a fixed-fee basis for the scope in Section 3, with milestone payments at the end of each phase in Section 5. Cloud consumption charges are billed directly to the customer's AWS account and are not included in the fee. Brightpath Cloud Consulting has assumed that the customer will provide timely access to its AWS account, to its PayFlow sandbox credentials and to subject matter experts for up to four hours per week. Change requests that alter the scope in Section 3 will be estimated and agreed in writing before work begins.

Support after the hypercare period is not included and may be contracted separately under a managed services agreement. Training for up to eight of the customer's staff is included.

7. Risks and Dependencies

The main delivery risk is the availability of the customer's engineering lead for design questions during the order-processing phase. The main technical dependency is the PayFlow sandbox, which must support the load tests planned for the cut-over phase. Both items will be reviewed in the weekly steering meeting and escalated to the sponsor where necessary.